Project Workflow

Mapping Invasive Alien Plant Species Across Virginia, United States

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Data/References

Available/Accessible Datasets

We have the boundary data of Virginia, and the United States. For the Land Use Land Cover datasets, we are unclear on how to incorporate it within R. We're considering loading in the data using ESRI API. On the other hand, the species distribution dataset has been one of the major hurdles. We are unable to find specific dataset for invasive plant species in Virginia. We've explored possible sources such as I Naturalist, Global Biodiversity Information Facility (GBIF), and the Virginia Department of Conservation and Recreation. For instance, GBIF contains plant species across Virginia. On filtering the data to invasive species, the search returned only one plant species, which does not satisfy our analysis. Wondering if we could tweak the study to exploring plant species that are (under the red list categories) exploring endangered, vulnerable or extinct species for example.

Data Name	Source	Description	Temporal Extent and Resolution	Spatial Extent and Resolution	Data Found
ESRI Land Use Land Cover	ESRI	Land use change	2017-2023	2m resolution	Yes
Virginia Boundary	ESRI Living Atlas	Study area boundary	--	--	yes
Temperature	NOAA	--	1851- Present	Global	No
Precipitation	NOAA	--	1851- Present	Global	No
Species Distribution Data	Virginia Department of Conservation and Recreation	--	--	Virginia	No

Workflow Diagram

Proposed Figures

Lan use type are critical factors in species habitats. Therefore, our result will illustrate the land use map in continental USA and Land Use map of Virginia

Temperature and Precipitation maps and plots

We shall also explore relationship between these variables and how they affect/ support invasive species.

Revised Research Questions and Likely Gaps in Knowledge

We're majorly doing a spatial analysis of invasive plant species distribution in Virginia using R. This analysis could be done using GIS software such as QGIS and ArcGIS, but this project will enable us to explore the capabilities of the R programming language for spatial analysis.

The conceptualization, writing, data handling and coding is equally down by both students and there is no conflict of interest.